

Investing Basics Workbook

Building your first portfolio -- a plain-language guide for someone about to open their first brokerage account.

This product contains educational information and templates. Nothing here is professional, legal, tax, or investment advice. Numbers in every worked example are illustrations of a method, not predictions or guarantees.

01 What you're actually buying

Almost everything you can buy in a brokerage account is one of three things, and understanding the difference between them will save you more money over your investing life than almost any single stock pick.

A stock

A stock is a share of ownership in one company. When you own a share of a company, you own a small slice of its future profits (or losses). If that one company does badly -- a product flops, a competitor eats its market, a scandal breaks -- your investment does badly with it, in direct proportion to how much of your money is in that one name. This is why holding only a handful of individual stocks is riskier than it looks on paper: you are betting heavily on a small number of specific outcomes, and being right about a company's products or leadership is genuinely hard to do consistently, even for professionals who do it full time.

An index fund

An index fund is a single fund that holds hundreds or thousands of companies at once, tracking a whole market or slice of it (the S&P 500 is the most common example, tracking roughly the 500 largest publicly traded U.S. companies). One purchase spreads your money across everything in that index automatically -- that spread is the entire point. If one company in the index collapses, it barely moves the needle on your total return, because it was never more than a tiny fraction of your holding to begin with. Index funds also tend to carry very low fees (often well under 0.10% per year) because there's no manager actively picking stocks -- the fund just mechanically holds what the index holds.

A bond

A bond is a loan you make to a government or company, which pays you interest on a schedule and returns your principal at a set maturity date. Bonds have historically been less volatile than stocks and have historically returned less over the long run. Their job in a portfolio isn't to make you rich -- it's to cushion the portfolio when stocks fall, since bond prices don't typically move in lockstep with stock prices.

02 Risk tolerance worksheet

Before choosing an allocation, be honest with yourself about how you'd actually behave in a downturn -- not how you'd like to imagine you'd behave. Answer each question below, then use your answers to read the allocation guidance in Section 3.

1. Your portfolio drops 25% in a single month. What do you actually do? (a) Sell most of it -- I can't watch that happen. (b) Sell some and wait it out. (c) Change nothing. (d) Buy more while it's cheap.
2. When will you actually need this money? (a) Within 3 years. (b) 3-10 years. (c) 10-20 years. (d) 20+ years, or retirement.
3. If you lost your income tomorrow, how many months of expenses do you have saved separately from this investment account? (a) Less than 1. (b) 1-3. (c) 3-6. (d) 6 or more.

4. Your experience with investing so far: (a) None -- this is my first account. (b) A retirement account I don't really manage. (c) I've invested through a full market drop before. (d) Comfortable, I follow it closely.

My answers: Q1 ____ Q2 ____ Q3 ____ Q4 ____

Score each answer a=1, b=2, c=3, d=4 and add the four numbers together. A total of 7 or under generally points toward a Conservative allocation; 8-11 toward Balanced; 12 and above toward Growth-oriented. These labels describe a starting point for the next section, not a fixed rule -- the honest answer to Question 1 usually matters more than the arithmetic.

03 Asset allocation by age and risk profile

A widely used starting point is the "110 minus your age" rule of thumb: subtract your age from 110 to get a rough stock percentage, with the rest split between bonds and a small cash cushion. It's a conversation starter, not a personalized recommendation -- your risk score from Section 2 should nudge it up or down.

RULE OF THUMB

Stock % = 110 - Age

Bond % = 100 - Stock % - Cash %

Cash % = 5 (typical), or 10 if your risk score was 7 or under

Worked example: age 25, Balanced risk score

Base stock % = 110 - 25 = 85%

Balanced score: no adjustment

Cash = 5% -> Bonds = 100 - 85 - 5 = 10%

Result: 85% stock / 10% bond / 5% cash

Worked example: age 50, Conservative risk score

Base stock % = 110 - 50 = 60%

Conservative score: subtract 15

Adjusted stock % = 60 - 15 = 45%

Cash = 10% (Conservative gets a larger cushion)

Bonds = 100 - 45 - 10 = 45%

Result: 45% stock / 45% bond / 10% cash

Notice that the same age can land in very different places depending on temperament and time horizon -- the age-only rule is a floor to start the conversation, not the finished answer.

04 Monthly contribution projector

Consistent monthly contributions compound over time in a way that's easy to state and genuinely counterintuitive to feel. Two formulas do all the work here: one for a lump sum growing on its own, and one for a stream of monthly contributions growing alongside it.

COMPOUND GROWTH FORMULAS

Future value of a lump sum:

$$FV = \text{Start} \times (1 + r)^n$$

Future value of monthly contributions:

$$FV = \text{Monthly} \times (((1 + r)^n - 1) / r)$$

where r = annual rate / 12 (monthly rate), n = number of months

Worked example: \$1,000 start + \$250/month at 7% annual return

Monthly rate $r = 0.07 / 12 = 0.005833$

After 10 years ($n = 120$ months):

Lump sum growth:	$1,000 \times (1.005833)^{120}$	=	\$2,009.66
Contribution growth:	$250 \times (((1.005833)^{120} - 1) / r)$	=	\$43,304.29
Total portfolio value:			\$45,313.95
Total contributed ($1,000 + 250 \times 120$):			\$31,000.00
Growth from compounding alone:			\$14,313.95

After 20 years ($n = 240$ months):

Total portfolio value:	\$137,972
Total contributed:	\$61,000
Growth from compounding alone:	\$76,972

The gap between years 10 and 20 is the entire lesson: contributions roughly double (from \$31,000 to \$61,000) but the portfolio more than triples. That acceleration is compounding doing its job, and it is the single strongest argument for starting early rather than waiting for a "better" moment.

Your own projection

Starting amount: \$_____ Monthly contribution: \$_____ Assumed annual return: _____ %

Value after 5 years: \$_____ Value after 10 years: \$_____

Value after 20 years: \$_____ Value after 30 years: \$_____

05 Before you open the account -- checklist

[] High-interest debt handled first

A credit card at 20-25% APR reliably costs you more than a diversified portfolio is likely to earn. Paying it down is the higher-certainty return -- there's no risk-adjusted argument for investing ahead of it.

[] Emergency fund exists, separately

If an unexpected bill forces you to sell investments during a downturn, you lock in the loss instead of riding it out. A cash reserve of 3-6 months of expenses, held outside the brokerage account, is what prevents that forced sale.

[] Employer retirement match captured

If your job matches 401(k) or similar contributions, that match is part of your compensation. Not contributing enough to get the full match is walking away from guaranteed money before you've invested a dollar of your own.

[] You know the fund's expense ratio

An expense ratio is the fund's annual fee as a percentage of your balance. The gap between a 0.03% fund and a 0.75% fund looks tiny on a statement but compounds into real money -- on \$50,000 over 20 years, that difference alone is worth tens of thousands of dollars.

[] A contribution schedule, not a timing plan

Deciding "\$X on the 1st of every month" removes the hardest and most repeated decision in investing -- when to buy. Trying to time the market adds risk and stress without a reliable payoff; a fixed schedule (often called dollar-cost averaging) sidesteps the question entirely.

06 Market sectors and sample ETFs

An index fund like a total-market or S&P 500 fund already owns companies across every sector -- that broad exposure is the whole point, and it's enough for most people, most of the time. Sector investing (buying a fund concentrated in one slice of the economy) is a more advanced, more concentrated choice: useful to understand, not a replacement for a diversified core holding. The market is commonly split into eleven sectors:

- Technology -- software, hardware, semiconductors. Example sector ETF: XLK.
- Health Care -- pharmaceuticals, biotech, medical devices, insurers. Example: XLV.
- Financials -- banks, insurance, asset managers. Example: XLF.
- Consumer Discretionary -- retail, autos, travel; spending that rises and falls with confidence. Example: XLY.
- Consumer Staples -- food, household goods; spending that holds up in a downturn. Example: XLP.
- Energy -- oil, gas, and related services. Example: XLE.
- Industrials -- manufacturing, aerospace, transportation. Example: XLI.
- Materials -- mining, chemicals, raw inputs for other industries. Example: XLB.
- Utilities -- electricity, water, gas providers; historically slower-moving and dividend-heavy. Example: XLU.
- Real Estate -- REITs holding commercial and residential property. Example: XLRE.
- Communication Services -- telecom, media, and major internet/social platforms. Example: XLC.

These tickers are the widely-tracked Select Sector SPDR funds, named here as real, concrete examples of what a sector fund looks like -- not a recommendation to buy any of them. Sectors move in and out of favor with the economic cycle (Energy and Financials tend to do relatively better when rates rise, for instance; Utilities and Staples tend to hold up better in a slowdown), which is exactly why concentrating in one is a bet on that cycle, not a bet on the market as a whole.

07 Dividends, explained

A dividend is a portion of a company's profit paid out directly to shareholders, typically on a quarterly schedule, typically as cash deposited into your brokerage account per share you own. Not every company pays one -- many younger or fast-growing companies reinvest all their profit back into the business instead, on the theory that growing the company is worth more to shareholders than a cash payout right now. Older, more established companies (utilities, consumer staples, many financials) are more likely to pay steady dividends because they have less need to reinvest every dollar of profit.

DIVIDEND YIELD

$\text{Dividend yield} = \text{Annual dividend per share} / \text{Current share price}$

Example: a stock paying \$2.00/year, trading at \$50/share
 $\text{Yield} = 2.00 / 50 = 4\%$

A dividend's declared amount doesn't change day to day, but its yield moves with the share price -- the same \$2.00 annual dividend is a 4% yield at \$50/share and a 2% yield at \$100/share. A very high yield can be a genuine opportunity or a warning sign (the market pricing in an expected dividend cut); it's a number to investigate, not to chase blindly.

- Ex-dividend date -- buy the stock on or after this date and you don't receive the upcoming payment; buy before it and you do.
- Dividend reinvestment (DRIP) -- most brokerages let you automatically use dividend cash to buy more shares of the same fund instead of holding it as cash, which is what most of the worked compound-growth examples in this workbook assume happens.
- Qualified vs. ordinary dividends -- qualified dividends (most U.S. stock dividends held over 60 days) are taxed at the lower long-term capital gains rate; ordinary dividends are taxed as regular income. This is a real distinction with real tax consequences -- confirm your own situation with a tax professional, not this workbook.

08 Retirement account types

Which account you invest through matters as much as what you invest in -- the account type determines when you pay tax, not just how much.

Traditional 401(k) / 403(b) (employer-sponsored)

Contributions come out of your paycheck before tax, lowering your taxable income today. The money grows tax-deferred, and you pay ordinary income tax when you withdraw it in retirement. Many employers match a percentage of what you contribute -- that match is free money and the reason this checklist puts it ahead of almost everything else. Contribution limits are set annually by the IRS and change most years -- check the current year's limit rather than relying on a number printed here.

Roth 401(k) (employer-sponsored)

The mirror image: contributions come out of your paycheck after tax, so there's no deduction today, but qualified withdrawals in retirement -- including all the growth -- are entirely tax-free. Increasingly common as an employer option alongside (not instead of) the traditional 401(k).

Traditional IRA (individual)

Opened on your own, not through an employer. Contributions may be tax-deductible depending on your income and whether you're also covered by a workplace plan; the account grows tax-deferred, and withdrawals in retirement are taxed as ordinary income, the same shape as a traditional 401(k) but without an employer match.

Roth IRA (individual)

Contributions are after-tax, and qualified withdrawals -- including growth -- are tax-free, same shape as a Roth 401(k). Roth IRAs have income limits on who can contribute directly, and generally smaller annual contribution limits than a 401(k). Many people use both: a 401(k) up to the employer match, then a Roth IRA for additional savings.

The core tradeoff

Traditional accounts save you tax now and cost you tax later; Roth accounts cost you nothing extra now (you already pay tax on that income either way) and save you tax later. Which is better depends on whether you expect your tax rate to be higher or lower in retirement than it is today -- a genuinely personal question, and one worth a real conversation with a tax professional rather than a rule of thumb. One thing that isn't personal: withdrawing from any of these accounts before age 59 1/2 typically triggers a 10% early-withdrawal penalty on top of ordinary tax, with a short list of specific exceptions -- treat retirement accounts as money you don't touch until retirement.

09 Quick reference

Allocation rule of thumb: Stock % = 110 - Age, adjusted by risk score. Compound growth: $FV = Start \times (1+r)^n$ for a lump sum, $FV = Monthly \times (((1+r)^n - 1)/r)$ for contributions, where r is the monthly rate. Dividend yield = annual dividend per share / share price. Traditional accounts: tax break now, taxed on withdrawal. Roth accounts: no break now, tax-free withdrawal. Before investing: emergency fund funded, high-interest debt cleared, employer match captured, expense ratios checked, contribution schedule set on autopilot.